

NATIONAL KALA-AZAR ELIMINATION PROGRAMME

STANDARD OPERATING PROCEDURE

for

Determining Place of Infection (Pol)

for Confirmed Kala-azar (Visceral Leishmaniasis) Cases
In India Based on Travel History

National Centre for Vector Borne Diseases Control (NCVBDC)
Ministry of Health & Family Welfare, Government of India

Abbreviations

CHC	Community Health Centre
CMO	Chief Medical Officer
DEC-KA	District Expert Committee on Kala-azar
DMO	District Malaria Officer
DVBDCO	District Vector Borne Disease Control Officer
ICMR	Indian Council of Medical Research
IDSP	Integrated Disease Surveillance Programme
IP	Incubation Period
IRS	Indoor Residual Spray
KA	Kala-azar
NCVBDC	National Centre for Vector Borne Diseases Control
NGO	Non-Governmental Organization
NVBDCP	National Vector Borne Disease Control Programme
PHC	Primary Health Centre
PKDL	Post Kala-azar Dermal Leishmaniasis
PoI	Place of Infection
RDT	Rapid Diagnostic Test
SEC-KA	State Expert Committee on Kala-azar
SOP	Standard Operating Procedure
SPO	State Programme Officer
UP	Uttar Pradesh
VBD	Vector Borne Disease
VL	Visceral Leishmaniasis
WHO	World Health Organization

PART I: ROUTINE KALA-AZAR CASES- PLACE OF INFECTION (PoI) DETERMINATION

1. Purpose and Scope

This SOP provides a standardized, evidence-based, and operationally feasible protocol for determining the probable **Place of Infection (PoI)** for confirmed Kala-azar (VL) cases in India. Accurate PoI determination is critical for:

- Classification of resident vs non-resident cases
- Identifying potential local transmission foci vs. imported cases
- Guiding targeted vector control interventions (e.g., IRS, environmental management)
- Informing accurate risk maps and programmatic priorities
- Strengthening surveillance and accelerating the goal of elimination (<1 case per 10,000 population at sub-district level)

This SOP is designed for use by Medical officers/peripheral health staff with clear algorithms and practical tools.

2. Definitions

- **Kala-azar (VL) Case:** Confirmed per national guidelines (clinical signs + rK39 rapid test and/or parasitological confirmation).
- **Probable Place of Infection (PoI):** The specific geographic location (ideally village/block/town) where the case was most likely infected by the bite of an infected sandfly, based on the integration of travel history and the parasite's incubation period.
- **Incubation Period:** The time between the infective sandfly bite and the onset of symptoms. For *L. donovani* in India: **typically, 2 to 6 months, but can range from 10 days to 2 years.** (Source: WHO Technical Report Series 949, 2022).
- **Residence:** The patient's usual place of dwelling.
- **Transmission Season:** Primarily post-monsoon and summer months (May-September, with peak around July-September in endemic areas of Bihar/Eastern Uttar Pradesh (UP)/Jharkhand/West Bengal), coinciding with high sand fly vector density.
- **Endemic Area/Village:** As notified under the National Kala-azar Elimination Programme (typically villages/blocks with reported cases in previous years). Four endemic states in India: Bihar, Jharkhand, West Bengal, and Eastern UP.

3. Roles and Responsibilities

- **Medical Officer (PHC/CHC):** Primary investigator. Responsible for conducting the PoI interview using the **Travel Timeline Calculator (Annex 1)**, completing the form, and making the preliminary PoI determination.
- **District Vector Borne Disease Control Officer/District Malaria Officer:** Validates PoI determinations, aggregates data, and ensures appropriate vector control responses in identified PoI.
- **State Programme Officer:** Monitors PoI data trends, resolves complex cases, and integrates findings into state-level planning.

- **WHO/Technical Partner:** Provides technical oversight, capacity building, and support for complex case analysis.

Important Notes for Medical Officers

- **Sensitivity:** Travel for migrant work is common. Assure the patient this information will not affect their treatment or cause any stigma.
- **Memory Aid:** Use local festivals, harvest seasons, or school dates to help patients recall travel dates.
- **Cross-Check:** Gently verify key travel dates with family members if possible.
- **No Blame:** The goal is to find the *geographical site of risk*, not to blame the patient or any community.
- **Update Maps:** Inform your district office if a case is assigned to a village not previously listed as endemic or if the case belongs to a village and block located out of district

4. Scientific and Operational Principles

- **Principle 1:**
In an endemic area, the patient's residence is the *default assumption* for Pol, unless travel history investigation provides compelling evidence to shift it elsewhere.
- **Principle 2:**
Infection occurs where the person is present during the transmission season, within the incubation period window prior to symptom onset.
- **Principle 3:**
Shorter stays (even overnight) in highly endemic villages during peak transmission season can pose significant risk, especially if sleeping conditions are conducive to sandfly bites (e.g., mud walls, cattle proximity, lack of bed nets).
- **Principle 4:**
The investigation is a structured interview, not an interrogation. Build rapport with the patient/family for accurate information.

5. Step-by-Step Procedure for Determining Place of Infection (Pol)

Step 1: Prepare for the Interview (At Case Confirmation)

- I. Confirm the VL diagnosis as per national guidelines.
- II. Note the **date of symptom onset: fever** (as accurately as possible). This is your anchor Point.
- III. Gather list and maps (if possible) of the district and known endemic villages/blocks.

Step 2: Conduct the Travel History Interview

- I. **Explain Purpose:** Briefly tell the patient/family, "To help prevent others from getting sick, we need to understand where you have stayed in the months before you fell ill."
- II. **Establish Residence:** Confirm the complete home address.
- III. **Use the Travel Timeline Calculator (Annex 1):**
 - ✓ Mark the **Date of Symptom Onset (fever)** (Box A).

- ✓ Calculate **6 months prior** (Box C - end of *typical incubation window*) and **2 months prior** (Box B - start of *typical incubation window*). These boxes define your **Critical Exposure Period**.
- ✓ For a case reported in non-endemic area, if there is no travel to an endemic area in 6 months prior to **date of onset of fever (symptoms)**, inquire about travel to endemic area in past 2 years and refer it to district and state expert committee

Systematic Enquiry: Ask about **all overnight stays** during the entire period from Box C to Box A. For each location:

- ✓ **Name of Village/Town/City.**
- ✓ **Dates of stay (From – To).**
- ✓ **Purpose of visit** (e.g., residing, migrant work, visiting relatives, tourism).
- ✓ **Sleeping conditions** (house type, use of bed net, presence of cattle).
- ✓ **Knowledge of KA/PKDL/fevers in that locality** during or after their stay.

Step 3: Analyze Information and Determine Pol

Use the **Pol Decision Algorithm (Annex 2):**

- I. **Did the patient travel or stay overnight outside their village of residence during the Critical Exposure Period (Box C to Box A)?**
 - ✓ **NO** → Probable Pol is the **Village of Residence**. Investigation complete.
 - ✓ **YES** → Proceed to Step 2.
- II. **Filter for Transmission Risk:**
 - ✓ List all locations stayed during the Critical Exposure Period.
 - ✓ **Highlight** any stays that overlapped with known local **Transmission Season** in that area AND were in **known endemic villages/blocks**.
 - ✓ **Prioritize** stays in **highly endemic villages** (as per district office listings).
- III. **Apply Incubation Period Logic:**
 - ✓ For each travel location, calculate the time between the **end of the stay** and the **date of symptom onset**.
 - ✓ If this time interval falls within the **2 to 6 months (typical incubation window)**, that location becomes a strong candidate for Pol.
 - ✓ Stays ending 10 days to 2 months before symptoms are also plausible, especially for immunosuppressed individuals.
 - ✓ For exceptional cases reported from non-endemic area, record travel history of 2 years (if any) and refer it to district and state expert committee
- IV. **Make the Final Determination:**
 - ✓ If only ONE location (including residence) has a plausible epidemiological link (endemic area + transmission season + correct incubation interval), assign it as the Pol.
 - ✓ If MULTIPLE locations are plausible, assign Pol based on **highest risk**: longest stay in the most endemic village during peak transmission season.
 - ✓ If information is inconclusive between residence and another location, default to **Residence as Pol**.
 - ✓ If the case is clearly travelled from a non-endemic to an endemic area (e.g., tourist), assign Pol as the visited endemic location.

Step 4: Documentation and Reporting

- I. Complete the **Place of Infection Investigation Form (Annex 3)** in duplicate.
- II. Clearly circle the final determined **Probable Place of Infection** (Village, Block, District).
- III. Submit one copy to the District Vector Borne Disease Control Office within **7 days** of case confirmation.
- IV. Retain one copy at the PHC/CHC for records.

Step 5: Action Trigger

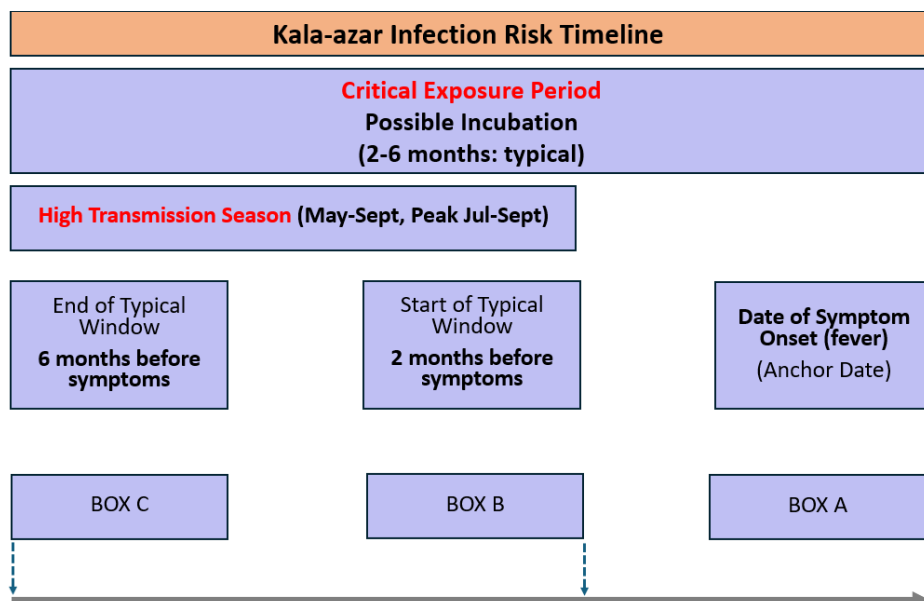
- The District Office must **cross-notify** the PHC of the determined POI (if different from case reporting PHC) within 3 days of receipt.
- The PHC in whose jurisdiction the POI falls must **initiate/strengthen active case search and vector control measures** in that village within one week.

References

1. WHO (2022). *WHO technical report series on control of leishmaniasis*. Geneva.
2. National Center for Vector Borne Diseases Control (NCVBDC), India. Kala-azar Elimination Programme guidelines.
3. Singh, O. P., et al. (2020). *Epidemiology of visceral leishmaniasis in India*. Journal of Tropical Medicine.
4. Stauch, A., et al. (2011). *Visceral leishmaniasis in the Indian subcontinent: Modelling epidemiology and control*. PLOS Neglected Tropical Diseases.
5. Operational studies from Bihar/Jharkhand highlighting migration and focal transmission.

ANNEX 1: Travel Timeline Calculator (Visual Aid for Interview):

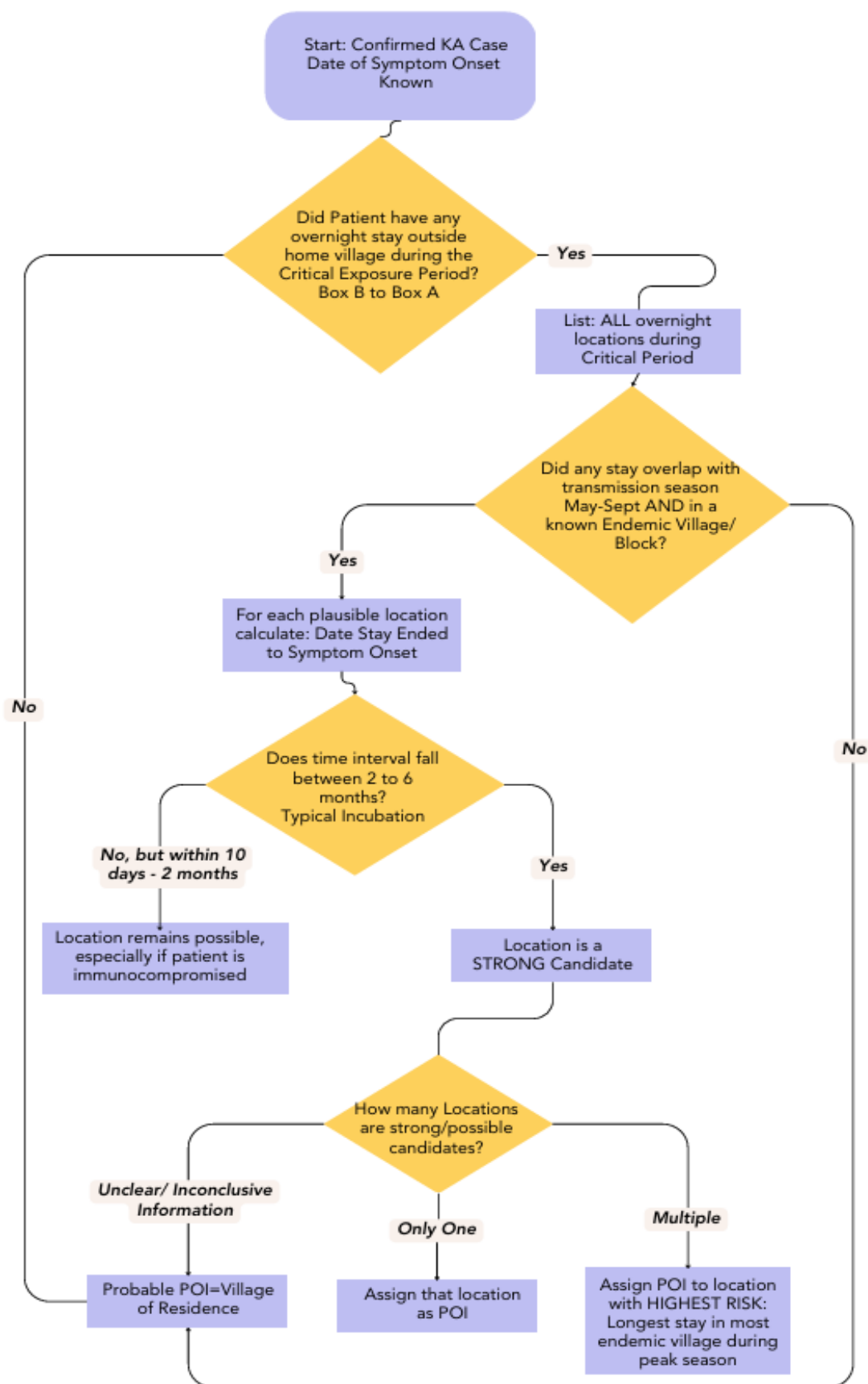
Instructions for Medical Officer: Use this chart during the patient interview. First, fill Box A with the date symptoms (fever) started. Then, calculate and fill Boxes B & C. Your goal is to investigate all overnight stays during the period from BOX C to BOX A.

**KA Case Travel Timeline: Critical Exposure Period to Investigate****Key Questions to Ask the Patient/Family:**

For the period FROM (BOX C) _____ TO (BOX A) _____:

1. "Did you sleep overnight in any village/town OTHER than your home village?"
2. "If YES, for each place, tell me:
 - ✓ Name of Place: _____
 - ✓ Dates Stayed (From/To): _____ to _____
 - ✓ Purpose of travel? (Work, visit family, etc.): _____
 - ✓ Where did you sleep? (Pucca/Kaccha house, used bed net? : _____
 - ✓ "Was that area known for Kala-azar or high fever/PKDL?: _____

**Note: The most likely Place of Infection (PoI) is often where the patient stayed during the Transmission Season, with the stay ending 2-6 months before Box A.*

ANNEX 2: Place of Infection (POI) Decision Algorithm (Flowchart):

ANNEX 3: Place of Infection (Pol) Investigation Form**Case Identification**

1. Patient ID: _____
2. Patient Name: _____
3. Reporting PHC: _____
4. Age (in Year) /Sex (M/F/Other): _____
5. Village of Residence: _____
6. Block of Residence: _____
7. District of Residence: _____
8. State of Residence: _____
9. Date of KA Confirmation: DD/MM/YYYY
10. Date of Symptom Onset (fever) (Box A): DD/MM/YYYY

Travel History (List ALL overnight stays from 6 months before symptoms to present)

11. Dates Stayed From: DD/MM/YYYY To: DD/MM/YYYY
12. Name of Village/Town, Block, District : _____
13. Purpose of Visit: _____
14. Sleeping Conditions: _____
15. Known KA in that area? (Y/N): _____

Place of Infection (Pol) Determination

16. Critical Exposure Period: From: DD/MM/YYYY To: DD/MM/YYYY
 17. Did travel occur in this period? (Yes/No): _____
 18. Locations overlapping Transmission Season in endemic areas(List): _____
 19. Calculated Incubation Intervals for plausible locations: _____
- Location: _____ Interval: _____
- Location: _____ Interval: _____
20. Probable Place of Infection (Pol) Determination:

Village: _____	District: _____
Block: _____	State: _____
 21. Justification for Pol Choice:
[e.g. "Only travel was to X during July, ended 4 months before symptoms." OR "No travel, Pol is residence."] _____
 22. Is Pol different from Patient's Residence? Yes / No

<i>If YES, Pol details for Cross-Notification:</i>	PHC of Pol: _____
District Office: _____	State Office: _____

23. Immediate Action Initiated at Reporting PHC: :_____

24. Case entered in KA Register: Yes / No Household/Contact screening initiated: Yes / No

25. DVBDC Office notified on: DD/MM/YYYY

26. Investigation Officer Name: _____ Designation: _____
Date: DD/MM/YYYY Signature: _____

For District Office Use

27. PoI Validated & Cross-Notified? : (Yes / NO) Date: DD/MM/YYYY

28. Remarks:_____

29. Signature of DVBDC Officer: _____

- ✓ Complete in duplicate, send one copy to District Vector Borne Disease Control Office within 7 days.
- ✓ Retain one copy at PHC.
- ✓ Use black/blue pen. Write clearly.

PART II: MANAGEMENT OF EXCEPTIONAL KALA-AZAR CASES WITH INDETERMINATE OR EXTENDED-INCUBATION PLACE OF INFECTION (POI)

1. Purpose and Scope

This SOP provides a standardized protocol for the investigation, referral, and expert committee review of confirmed Visceral Leishmaniasis (Kala-azar) cases where the probable Place of Infection (PoI) cannot be determined through the routine investigation protocol due to:

- Cases involving stay in multiple endemic areas or travel history extending beyond the typical 6-month incubation window (up to 2 years prior to symptom onset)
- Inconclusive or conflicting evidence making PoI assignment indeterminate.
- Cases requiring inter-district or inter-state transfer classification.
- Accurate resolution of these exceptional cases is critical for maintaining the integrity of epidemiological data, ensuring appropriate targeting of interventions, and upholding the credibility of elimination certification.

2. Definitions

- **Indeterminate Case:** A confirmed VL case where the PoI cannot be assigned per the routine SOP algorithm (Annex 2 of the main PoI SOP) due to extended or unclear travel history/exposure or any other reason.
- **Extended Incubation Window:** The period from **6 months to 2 years** prior to symptom onset, within which infection could have plausibly occurred.
- **Habitual/Permanent Residence:** The location of the patient's primary dwelling and social ties, supported by documentary or community evidence.
- **District Expert Committee on Kala-azar (DEC-KA):** A multidisciplinary district-level committee convened to adjudicate on exceptional PoI cases and inter-block transfers.
- **State Expert Committee on Kala-azar (SEC-KA):** A multidisciplinary state-level committee convened to adjudicate complex cases involving inter-district transfers.

Composition of District Expert Committee-KA (DEC-KA):

- Civil Surgeon / Chief Medical Officer (Chair)
- District Malaria Officer (DMO) / District Vector Borne Disease Control Officer (DVBD CO)
- District Epidemiologist or IDSP Officer
- District Entomologist
- Medical Specialist / Physician from District Hospital (preferably with VL experience)
- Entomologist or additional invitees (as required): representative from medical college, NGO/technical partner

Composition of State Expert Committee-KA (SEC-KA):

- State Program Officer-VBD or Kala-azar (Chair)
- State VBD Consultant
- State Entomologist
- State Epidemiologist or IDSP Officer

- Medical Specialist / Physician from Medical College (preferably with VL experience)
- Additional invitees (as required): ICMR, NGO/technical partner

3. Roles and Responsibilities

- **Medical Officer (PHC/CHC):** Identifies exceptional cases during initial investigation. Documents **full 2-year travel history**, compiles all evidence, and completes the **Exceptional Case Referral Dossier (Annex A)** for submission to the District Malaria Officer/DVBDCO.
- **Entomologist:** Provides entomological evidence on vector species, vector density trends, breeding sites and transmission patterns to support accurate assignment of probable place of infection.
- **District Vector Borne Disease Control Officer (DVBDCO):** Validates the referral dossier, convenes the DEC-KA, implements committee decisions, and oversees case transfer protocols.
- **District Expert Committee (DEC-KA):** Reviews evidence, makes evidence-based final decisions on Pol for exceptional cases and inter-block transfers, and documents justifications.
- **State Programme Officer (SPO - KA):** Reviews DEC-KA recommendations for inter-district cases, convenes the SEC-KA for unresolved or cross-border cases, and ensures state-level reporting integrity.
- **State Expert Committee (SEC-KA):** Makes final decisions on inter-district and complex cross-border Pol assignments. Provides guidance on cases with significant programmatic implications.
- **NCVBDC:** Serves as the final authority for guidance on inter-state/cross-border cases and maintains national-level data integrity.

4. Scientific and Operational Principles

- **Principle 1:** For cases with no travel to endemic areas within the typical 2-6 month window or in cases with stays in multiple endemic areas, investigation must be expanded to cover the full 2-year extended incubation period prior to symptom onset.
- **Principle 2:** The default assumption remains the **Habitual Residence**, unless strong, triangulated epidemiological evidence points to an alternative location.
- **Principle 3:** "Strong evidence" includes: a) Even one night stay to prolonged stay in an endemic focus during transmission season, b) supported by known local VL/PKDL transmission or vector presence, c) with a temporally plausible incubation interval.
- **Principle 4:** All decisions must be **evidence-based, transparently documented**, and made by a multidisciplinary expert committee to minimize bias.

5. Step-by-Step Procedure

Step 1: Identification and Documentation at PHC/CHC

1. During the initial Pol investigation, if the patient has **NO travel history to endemic areas within the 6-month critical window**, immediately expand the interview to cover the **past 2 years**.
2. Meticulously document all overnight stays, purposes, durations, and sleeping conditions for this 24-month period.
3. If the Pol remains indeterminate after this expanded history (e.g., multiple plausible sites, conflicting information), classify the case as **"Exceptional - For Committee Review"**.
4. Complete the **Exceptional Case Referral Dossier (Annex A)**, attaching all treatment records, lab reports, and the detailed travel timeline.

Step 2: Submission and Review at District Level

1. The Medical Officer submits the dossier to the DVBDCO within 7 days of case confirmation.
2. The DVBDCO reviews for completeness and convenes the **District Expert Committee (DEC-KA)** within 14 days of receipt.
3. The DEC-KA reviews the dossier using the **Evidence Assessment & Decision Matrix (Annex B)**.
4. **DEC-KA Decision Authority:**
 - ✓ Final determination of Pol for exceptional cases within the district.
 - ✓ Approval for **inter-block transfers** of residence or Pol.

Step 3: Committee Decision-Making & Documentation

1. The DEC-KA/SEC-KA shall deliberate based on the pre-defined matrix, prioritizing epidemiological plausibility.
2. The committee must record:
 - ✓ Final Pol assignment (Village, Block, District, State).
 - ✓ Justification citing specific evidence used (travel dates, endemicity status, incubation interval).
 - ✓ Any dissenting opinions.
3. These minutes must be signed by the chairperson and archived formally.

Step 4: Case Transfer Protocol (If Applicable)

For committee-approved transfers (inter-block or inter-district or inter-state), the following actions are **mandatory**:

1. **Data Management:** The originating block marks the case as "**Transferred Out**" in all registers (Laboratory, Treatment, KA Register). The receiving block enters it as "**Transferred In**".
2. **Formal Notification:** The DVBDCO issues a formal **Case Transfer Intimation Letter (Annex C)** to the DVBDCO of the receiving district/block, with copies to the respective SPO and originating PHC.
3. **Documentation Sharing:** A copy of the complete treatment card and investigation dossier is shared with the receiving PHC to ensure continuity of care and accurate follow-up (e.g., PKDL monitoring).
4. **State-Level Transfer:** For inter-district transfers, the DEC-KA recommendation is sent to the SPO for SEC-KA ratification. The SEC-KA issues the final transfer directive.
5. **Inter-State/Cross-Border Transfer:** The State Programme, endorsed by SEC-KA, formally refers the case to the NCVBDC via prescribed channels, submitting the entire dossier for national-level coordination.

ANNEX A: Exceptional Case Referral Dossier Template (To be completed by the Investigating Medical Officer)

Section A: Case Identifiers

1. Patient ID: _____
2. Patient Name: _____
3. Reporting CHC/PHC: _____
4. Age (in Year) /Sex (M/F/Other): _____
5. Date of Symptom Onset (fever): DD/MM/YYYY
6. Date of KA Confirmation: DD/MM/YYYY

Section B: Residence & Travel History (24-Month Timeline)

7. Habitual Residence (with supporting evidence: Aadhar Card, Ration card, Voter ID, etc.): _____
8. Name of Village/Town, Block, District : _____
9. Chronological List of ALL Overnight Stays (Past 24 Months):

Date		Location (Village/Town, Block, District, State)	Purpose	Sleeping Conditions	Know KA/PKDL in area? (Y/N)
From	To				

Section C: Reason for Referral

- No travel to endemic area in 6-month window; history extends to ____ months/years.
- Multiple plausible PoI locations within incubation window.
- Conflicting or unverifiable travel history.
- Request for inter-block/district/state transfer due to PoI determination.
- Other (Specify): _____

Section D: Investigating Officer Declaration

I confirm this dossier contains a complete account of the patient's travel history to the best of my ability.

Name: _____ Designation: _____

Date: DD/MM/YYYY

Signature: _____

ANNEX B: Evidence Assessment & Decision Matrix for Expert Committees

Evidence Category	Strong Evidence for Habitual Residence Pol	Strong Evidence for Alternate Location Pol	Indeterminate / Weak Evidence
1. Residence Proof	Documented permanent residence (ID, family). No alternative long-term base.	Habitual residence is non-endemic. Alternate location is a documented endemic focus.	Conflicting or lack of residence documents.
2. Travel History	No stays in other endemic areas during the full 2-year incubation window.	Overnight to prolonged stay in an endemic block during a Transmission Season , within a plausible incubation interval (10 days - 2 years).	Multiple short stays in various endemic areas. Dates are unclear or unverified.
3. Local Transmission Evidence	History of local VL/PKDL cases, vector presence, or IRS activities in habitual residence area during exposure period.	Verified local transmission (recent VL/PKDL cases, vector density data) in the alternate location during the patient's stay.	No data on transmission from either location.
Committee Decision Guide	Assign Pol to Habitual Residence.	Assign Pol to Alternate Location. Initiate transfer protocol.	Default to Habitual Residence. Strengthen surveillance in all potential exposure sites.

ANNEX C: Case Transfer Intimation Letter Template

[Official Letterhead of Originating District DVBDco]

To,

The DVBDco / CMO,

[Name of Receiving District/Block],

[State].

Subject: Intimation of Inter-District/Block Transfer of Kala-azar Case - [Patient ID/Name]

Dear Sir/Madam,

This is to formally intimate that the following confirmed Kala-azar case, investigated in our district, has been assigned a Place of Infection (PoI) within your jurisdiction as per the decision of the [DEC-KA/SEC-KA] on [Date].

Case Details:

1. Patient Name/ID:
2. Age/Sex:
3. Date of Symptom Onset:
4. Date of Diagnosis & Treatment Initiated:
5. **Assigned PoI:** Village _____, Block _____, District _____.

Reason for Transfer/PoI Assignment: [Brief justification from committee minutes].

The complete treatment and investigation dossier is attached for your records. Kindly ensure:

1. Entry of the case as a "**Transferred In**" case in your relevant registers.
2. Initiation of active case search and vector control measures in the assigned PoI village.
3. Continuity of care and PKDL follow-up as per guidelines.

A copy of this intimation is marked to the State Programme Officer for information.

Sincerely,

Signature

District Vector Borne Disease Control Officer

[Originating District Name]

Copy to:

State Programme Officer (Kala-azar), [State Name]